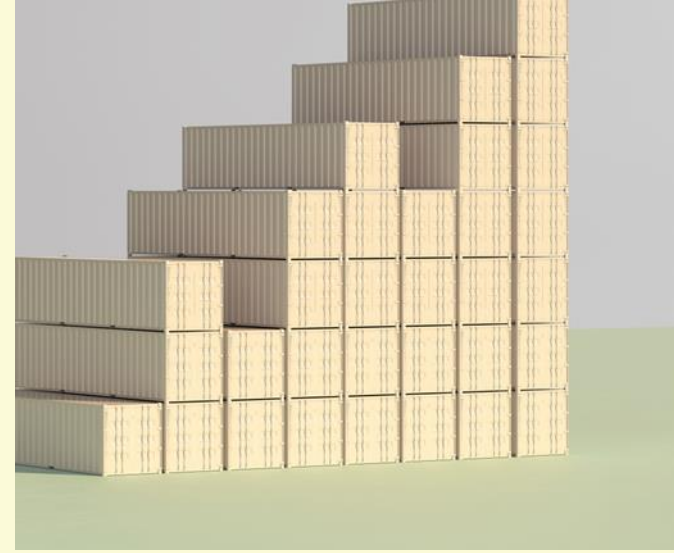




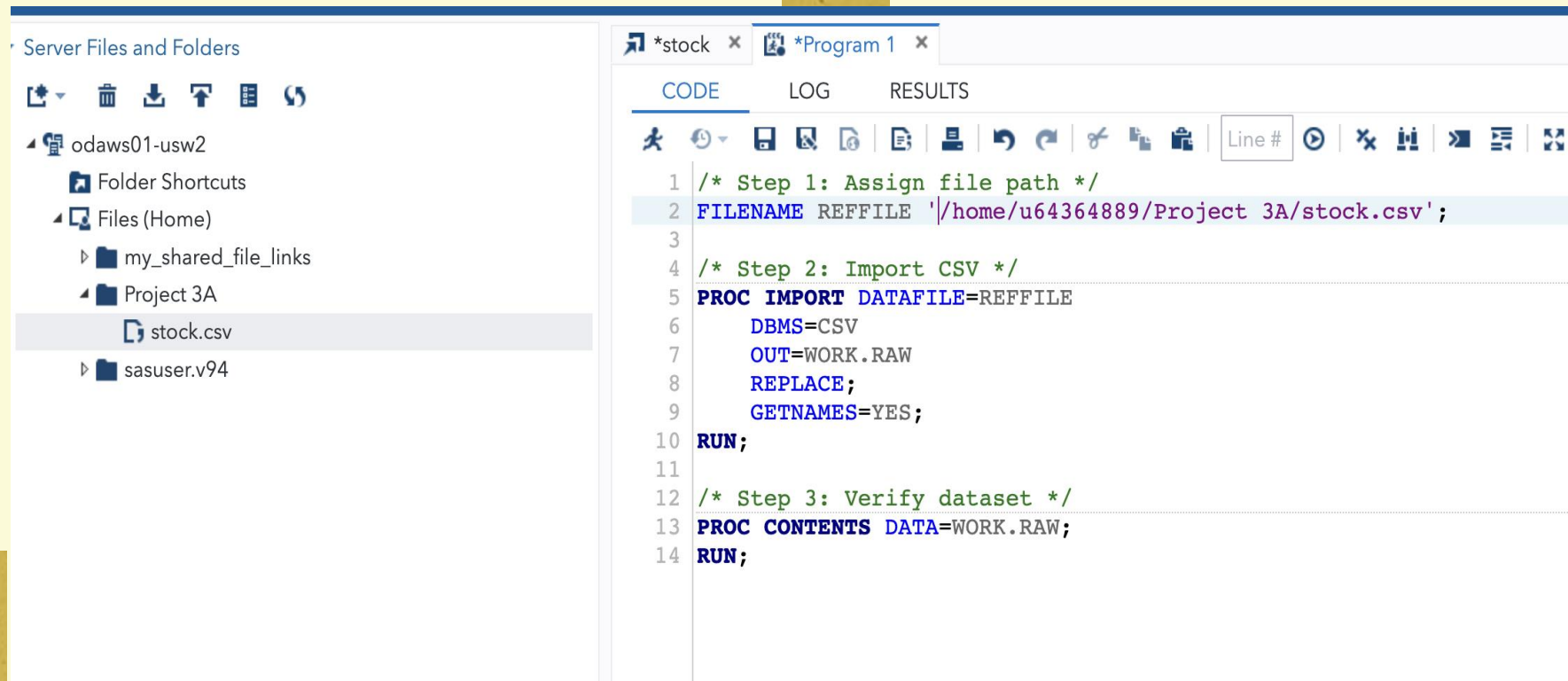
PROJECT 3A: VISUALIZATION OF TIME SERIES DATA

Data Analytics for FinTech
IDC4252C-2032
Karah Culley

Data Preparation

- The stock.csv file was uploaded into SAS Studio and imported into a WORK dataset
- The data was verified using PROC CONTENTS to ensure all variables were properly loaded

The output:

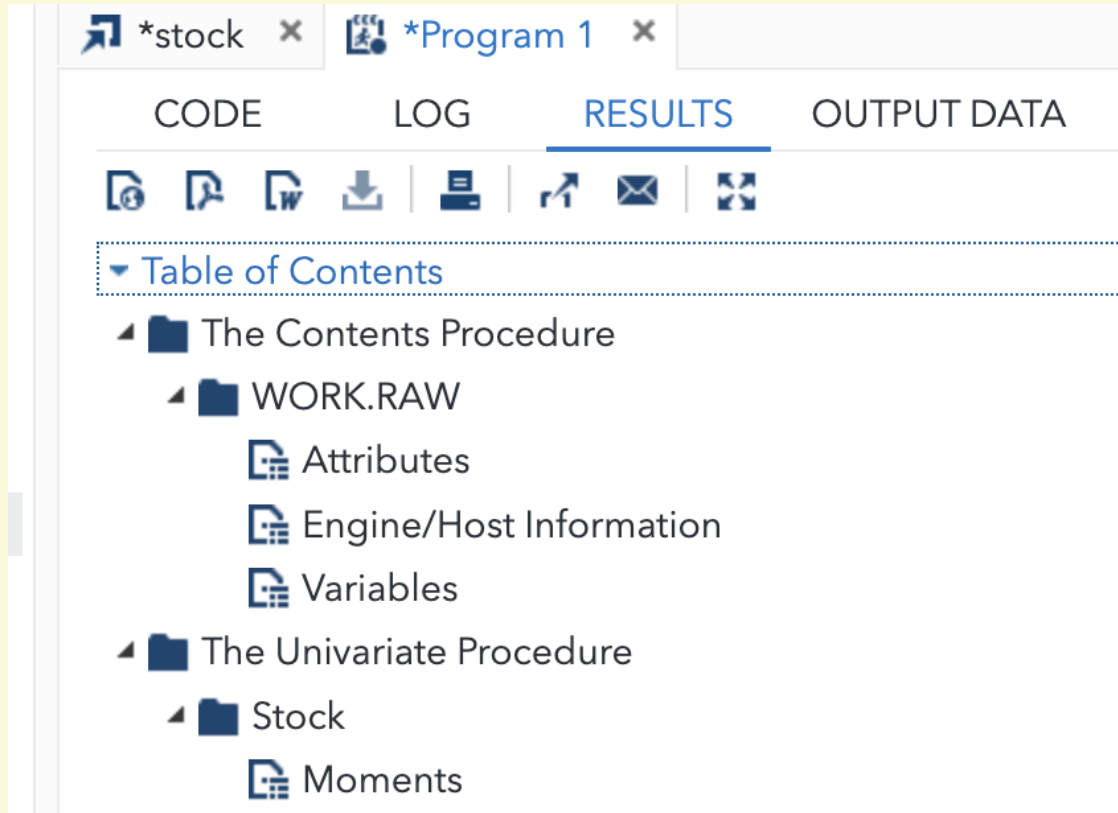


The screenshot displays the SAS Studio web interface. On the left, the 'Server Files and Folders' pane shows a tree structure with 'odaws01-usw2' as the root. Under 'Files (Home)', the 'Project 3A' folder is expanded, revealing the 'stock.csv' file, which is currently selected. The main workspace on the right is divided into two tabs: '*stock' and '*Program 1'. The '*Program 1' tab is active, showing a code editor with three steps of SAS code. The code includes comments for each step, file path assignment, data import using PROC IMPORT, and dataset verification using PROC CONTENTS. The interface includes standard file management icons at the top of the left pane and a toolbar with execution and editing icons above the code editor.

```
1 /* Step 1: Assign file path */
2 FILENAME REFFILE '/home/u64364889/Project 3A/stock.csv';
3
4 /* Step 2: Import CSV */
5 PROC IMPORT DATAFILE=REFFILE
6     DBMS=CSV
7     OUT=WORK.RAW
8     REPLACE;
9     GETNAMES=YES;
10 RUN;
11
12 /* Step 3: Verify dataset */
13 PROC CONTENTS DATA=WORK.RAW;
14 RUN;
```

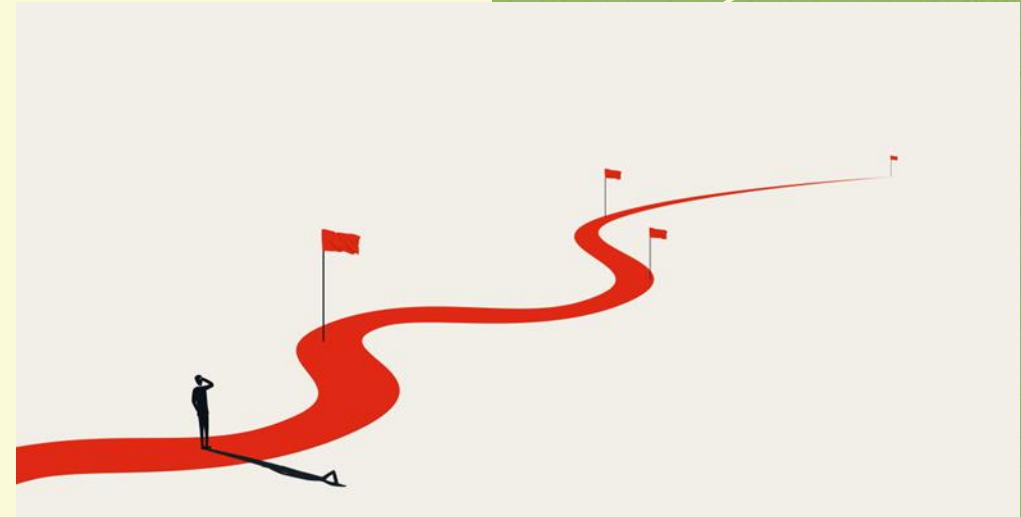
Table of Contents

This is the table of contents created for the SAS project:



The screenshot shows the SAS software interface with two tabs: '*stock' and '*Program 1'. The 'RESULTS' tab is selected, displaying a table of contents. The table of contents is organized into a hierarchical structure with folders and documents. The folders are 'The Contents Procedure', 'WORK.RAW', 'The Univariate Procedure', and 'Stock'. The documents are 'Attributes', 'Engine/Host Information', 'Variables', and 'Moments'.

CODE	LOG	RESULTS	OUTPUT DATA
Table of Contents			
The Contents Procedure			
WORK.RAW			
Attributes			
Engine/Host Information			
Variables			
The Univariate Procedure			
Stock			
Moments			



Descriptive Statistics

The output:

- PROC UNIVARIATE was used to summarize the stock variable

The output provides the measures:

- Mean
 - Standard deviation
 - Minimum and maximum values
 - Quartiles
- These statistics help describe the overall distribution of stock prices

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	180.3643	Pr > t	<.0001
Sign	M	297	Pr >= M	<.0001
Signed Rank	S	88357.5	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	5.83
99%	5.81
95%	5.73
90%	5.63
75% Q3	5.44
50% Median	4.71
25% Q1	4.42
10%	4.07
5%	3.48
1%	3.44
0% Min	0.37

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
0.37	203	5.81	585
3.43	30	5.81	588
3.43	27	5.82	586
3.43	24	5.82	589
3.43	23	5.83	587

The UNIVARIATE Procedure
Variable: Stock

Moments			
N	594	Sum Weights	594
Mean	4.88314815	Sum Observations	2900.59
Std Deviation	0.65984593	Variance	0.43539665
Skewness	-0.8858098	Kurtosis	2.62474774
Uncorrected SS	14422.2009	Corrected SS	258.190213
Coeff Variation	13.5127157	Std Error Mean	0.02707381

Basic Statistical Measures			
Location		Variability	
Mean	4.883148	Std Deviation	0.65985
Median	4.710000	Variance	0.43540
Mode	4.270000	Range	5.46000
		Interquartile Range	1.02000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	180.3643	Pr > t	<.0001
Sign	M	297	Pr >= M	<.0001
Signed Rank	S	88357.5	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	5.83
99%	5.81
95%	5.73
90%	5.63
75% Q3	5.44

Summary Statistics

- PROC MEANS was used to calculate:
 - Number of observations
 - Mean
 - Standard deviation
 - Minimum and maximum
- This confirms the central tendency and spread of the stock prices

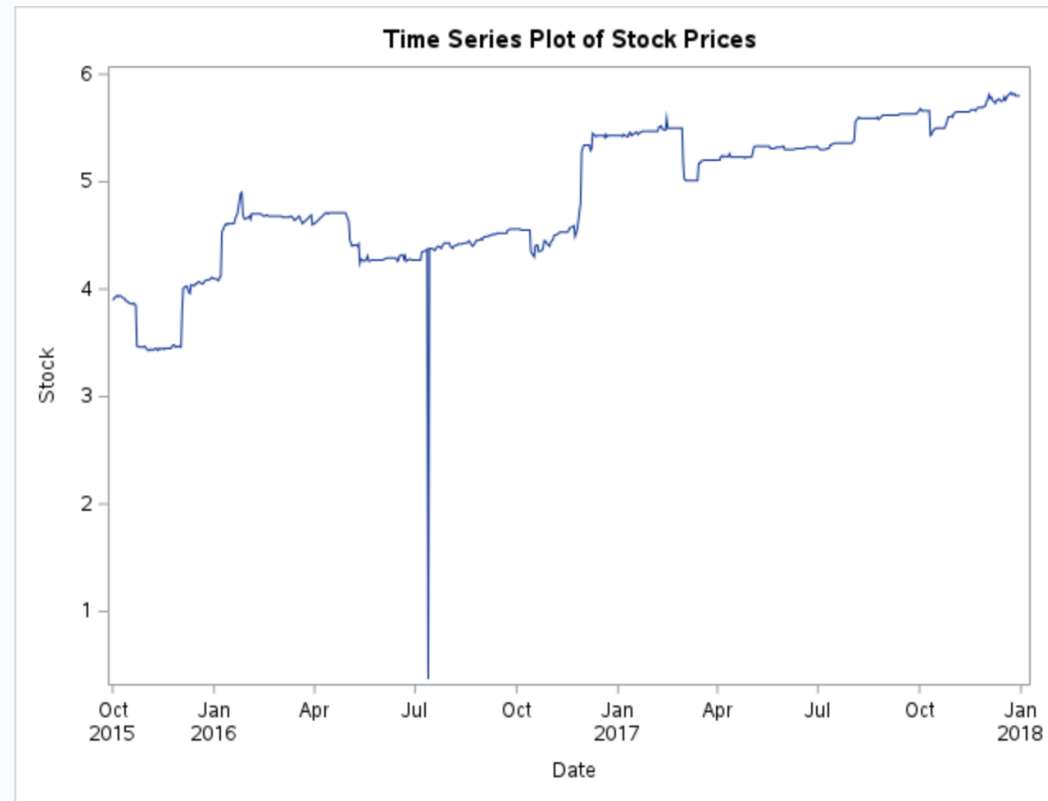
The output:

The MEANS Procedure

Analysis Variable : Stock				
N	Mean	Std Dev	Minimum	Maximum
594	4.8831481	0.6598459	0.3700000	5.8300000

Time Series Plot

The output:



- A time series plot was created to show how the stock price changes over time

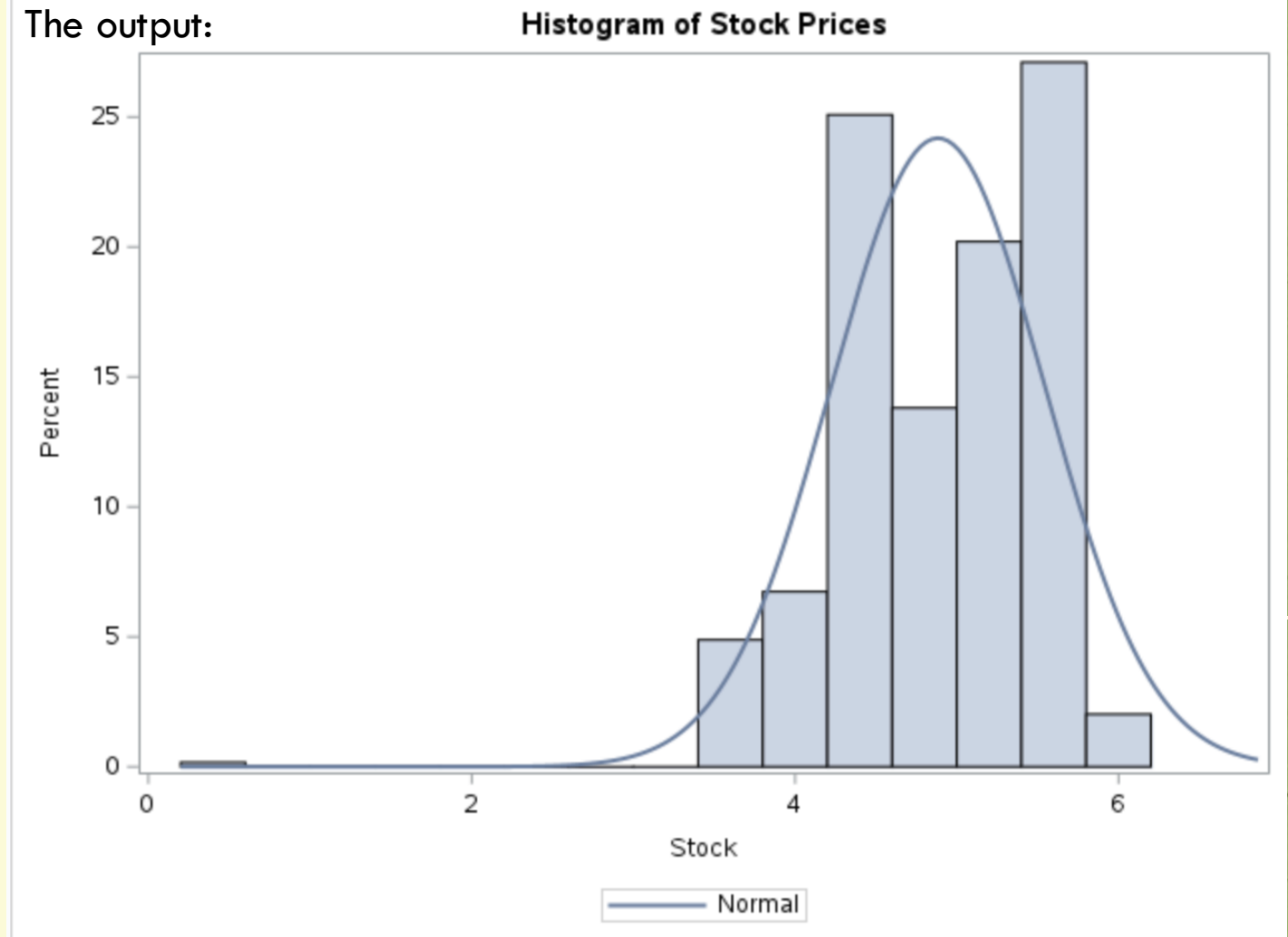
- The graph helps identify:
 - Overall trends
 - Periods of growth or decline
 - Volatility in the stock price





- A histogram was created to show the distribution of stock prices
- The graph shows:
 - How frequently different price levels occur
 - The overall shape of the distribution

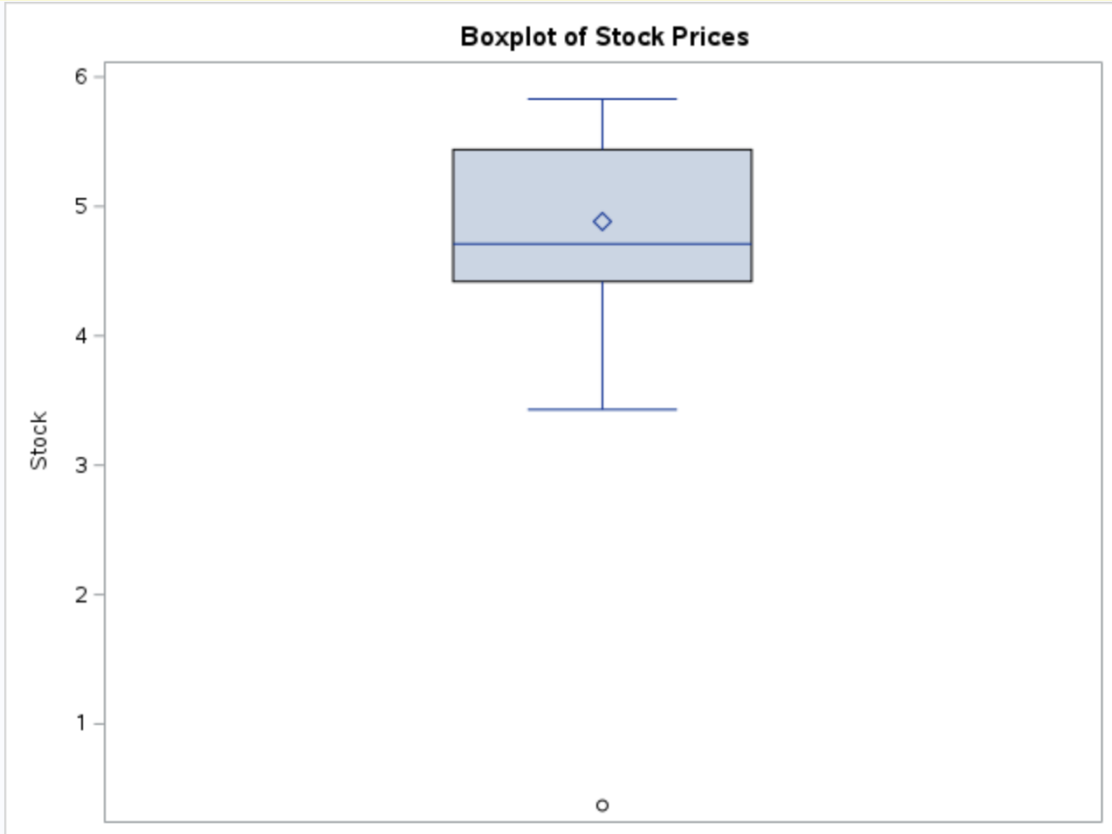
The output:



DISTRIBUTION OF STOCK PRICES

BOXPLOT ANALYSIS

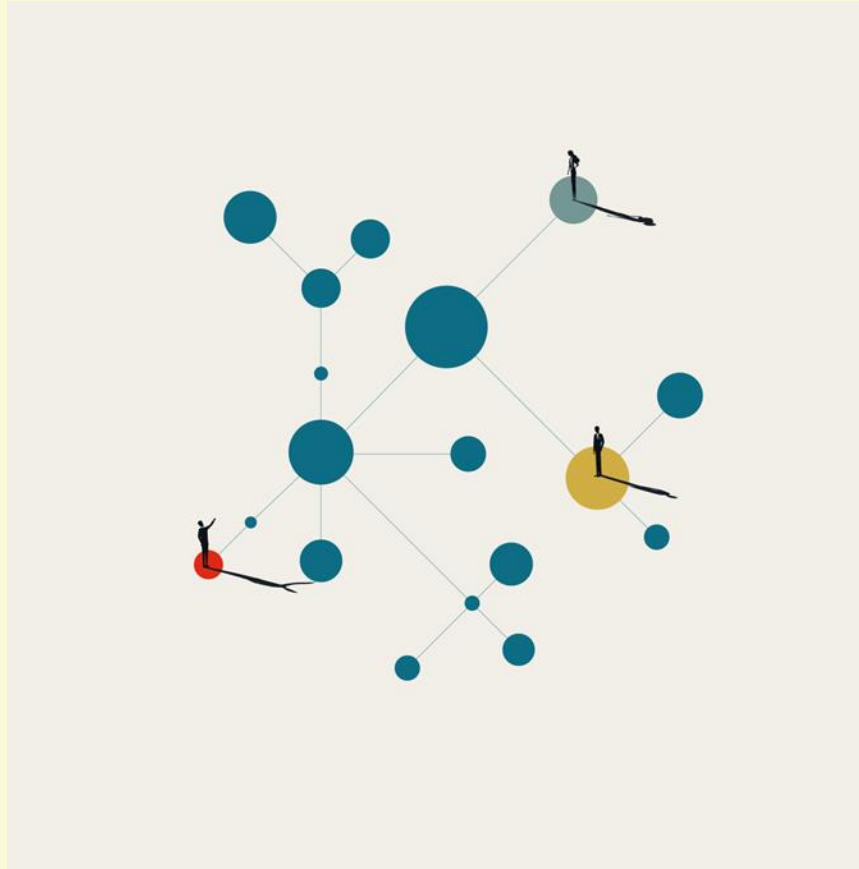
The output:



- A boxplot was used to visualize the spread of the stock prices.

- The boxplot shows:

- The median
- Interquartile range
- Possible outliers



THANK YOU
FOR YOUR
TIME, DR.
AHMETI

